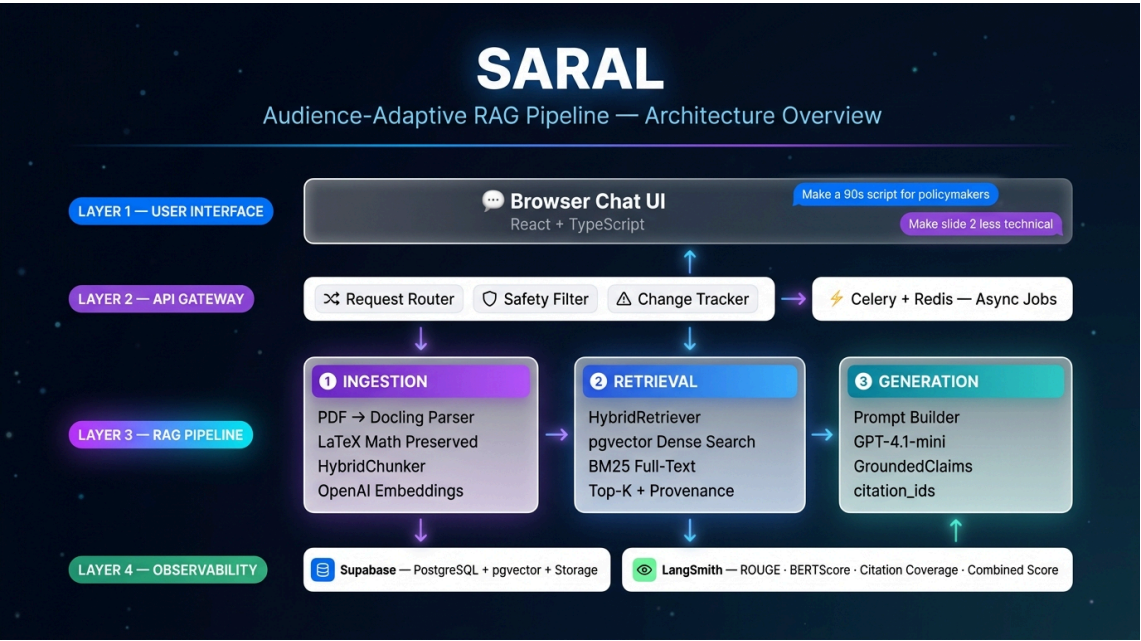


SARAL RAG Pipeline & Audience-Adaptive Script Generator Plan

1. Architecture Overview

The SARAL system utilizes a 4-layer architecture to handle complex document processing, retrieval, and audience-adaptive generation:

- **User Layer:** React/TypeScript UI for PDF uploads and iterative natural language edits.
- **API & Queue Layer:** FastAPI backend that dispatches long-running parsing and generation jobs to a Celery/Redis queue.
- **RAG Pipeline (Core):**
 - **Ingestion:** Docling parser that retains structural hierarchy and LaTeX math.
 - **Retrieval:** Hybrid pgvector + BM25 search for finding exact facts and mathematical formulas.
 - **Generation:** LLM Prompt Builder parameterized by Audience, Length, and Style.
- **Storage & Observability:** Supabase for vector storage and LangSmith for real-time evaluation (ROUGE-L, BERTScore).



2. Retrieval Index Construction & Chunking Strategy

Chunking Strategy (Preserving LaTeX Math)

To ensure that mathematical integrity is maintained, the chunking strategy relies on structural boundaries rather than arbitrary token counts:

- **Docling Native Parsing:** Extracts documents into semantic blocks (paragraphs, tables, equations).
- **Block-Level Chunking:** Chunks are created by grouping semantic blocks up to a token limit, but a math block ($...$ or $\begin{equation}...\end{equation}$) is **never split**.

- **Metadata Tagging:** Each chunk retains metadata such as `page_numbers` , `heading` , and a unique `chunk_id` .

Index Construction

- **Hybrid Search Index:** We index chunks in Supabase using `pgvector` (for semantic similarity of concepts) and `BM25` (for exact keyword and formula matching).
- **Reciprocal Rank Fusion (RRF):** Results from both vector and keyword searches are merged to retrieve the most relevant evidence, particularly useful for isolating exact equations.

3. Formula Parsing Pipeline

Formula parsing is handled rigorously at every step:

- **Ingestion:** Docling accurately identifies math elements and outputs them as raw LaTeX without dropping symbols.
- **Retrieval:** Both dense embeddings and sparse BM25 representations retain LaTeX syntax, allowing queries for specific mathematical variables (e.g., `\mathbf{W}`).
- **Generation (Prompt Builder):** The `grounding_rules` enforce strict mathematical fidelity. The LLM is instructed to:
 - **Never simplify or alter** LaTeX macros (e.g., copy `\mathbf{W}` exactly).
 - Wrap all mathematical formulas in standard markdown math delimiters (`$$` for blocks, `$` for inline).
 - Place citations *before* block equations to prevent rendering breaks.
 - **Double-escape** all LaTeX backslashes (`\\alpha`) to prevent JSON parsing crashes during structured output generation.

4. Intent Analysis & Iterative Edits

When a user submits a natural language edit, the system performs **Intent Analysis** to update the generation parameters:

- **Parameters:**
 - `{audience}` (policymakers, grad students, press)
 - `{length}` (30s, 90s, 5min)
 - `{style}` (technical, plain-English, press release)
 - `{change_instruction}` (e.g., "focus more on the empirical results")

The system parses the user's request, classifies the intent, and maps it to the nearest parameter constraints, injecting the explicit `{change_instruction}` into the prompt builder for the revision.

5. Prompt Template Family & Prompt Builder

The Prompt Builder dynamically constructs the context using validated `request` , `evidence` , and `grounding_rules` blocks. The core generation instruction is parameterized as follows:

```
You are an expert technical communicator writing a {length} script for {audience}.
The tone should be {style}.

{change_instruction}
```

Use the provided <evidence_pack> to ground every claim.

Instantiated Example 1 (The Policymaker Pitch)

- **{audience}**: Policymakers
- **{length}**: 90s
- **{style}**: plain-English
- **{change_instruction}**: "Emphasize the economic impact and remove complex derivations."
- **Resulting Prompt Context**: "You are an expert technical communicator writing a 90s script for Policymakers. The tone should be plain-English. Emphasize the economic impact and remove complex derivations. Use the provided ..."

Instantiated Example 2 (The Deep Dive)

- **{audience}**: Grad students
- **{length}**: 5min
- **{style}**: technical
- **{change_instruction}**: "Include the full objective function and explain the optimization constraints."
- **Resulting Prompt Context**: "You are an expert technical communicator writing a 5min script for Grad students. The tone should be technical. Include the full objective function and explain the optimization constraints. Use the provided ..."